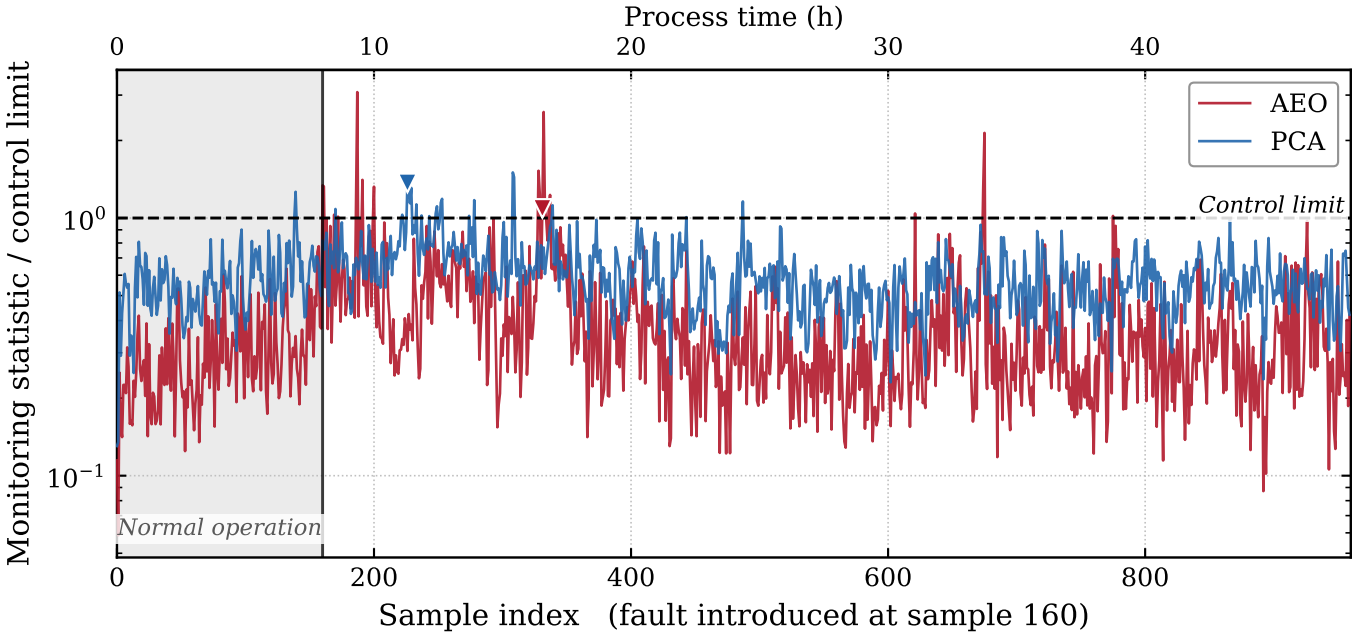
Fault Detection Performance for IDV(15)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay 344.0 samples (1032 min)	MDR 0.986	detected in 2/7 runs
PCA	delay 66.0 samples (198 min)	MDR 0.986	detected in 1/7 runs

Fault not reliably detected by either method (caught in fewer than half the runs)